Truth lemma clause for the box: necessity as agreement over all maximally accessible worlds

The Open Logic Project contributors

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Open Logic Project unit file sources/olp-normal-modal-logic-completeness/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: this proposition lives in the source's Box-primitive branch; the span is the branch's own lines verbatim; the diamond counterpart has its own card.

Editorial note. The label prop:box names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own prop environment in the unit source.

Allowed standard prerequisites (editorial).

- Box1: necessitation lifts to boxed premise sets. Same section, \ollabel{lem:box1} (source line 1155). Consumed one step earlier inside the cited box2 lemma.
- Box2: inverse-image derivability transfers through the box. Same section, \ollabel{lem:box2} (source line 1170, own card). Line 1245: from the failure of Box A at Gamma to the failure of A at the inverse-image set.
- Lindenbaum's lemma. Same part, \ollabel{thm:lindenbaum} (own card, line 1027). Line 1248: completing the consistent inverse-image-plus-negation set to a full world Delta.
- Consistency existence fact for adding a negation. Proof theory part, prop:consistencyfacts item (c). Line 1247: certifies the set fed to Lindenbaum is Sigma-consistent.

Source reference. Original lemma lem-box2: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1214–1214. Prerequisite pointer invoked at proof line(s) 1245; it names a standard fact of the same unit that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-consistencyfacts: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1247–1247. Named citation invoked at source line(s) 1247: the author's marker \olref[prf][con]{prop:consistencyfacts} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original proposition prop-consistencyfacts-b: sources/olp-normal-modal-logic-completeness/pinned-original-source.tex, source lines 1247–1247. Named citation invoked at source line(s) 1247: the author's marker \olref[prf][con]{prop:consistencyfacts-b} resolves inside the full Open Logic Project to another unit (not staged in this corpus); the cited fact is used as a standard result and every step of the counted proof built on it is retained verbatim in the delivered body.

Proposition 0.1. *If Γ is complete Σ-consistent, then $\Box A \in \Gamma$ if and only if for every complete Σ-consistent Δ such that $\Box^{-1}\Gamma \subseteq \Delta$, it holds that $A \in \Delta$.*

Proof. Suppose Γ is complete Σ-consistent. The "only if" direction is easy: Suppose $\Box A \in \Gamma$ and that $\Box^{-1}\Gamma \subseteq \Delta$. Since $\Box A \in \Gamma$, $A \in \Box^{-1}\Gamma \subseteq \Delta$, so $A \in \Delta$.

For the "if" direction, we prove the contrapositive: Suppose $\Box A \notin \Gamma$. Since Γ is complete Σ-consistent, it is deductively closed, and hence $\Gamma \vdash /[\Sigma]\Box A$. By lem-box2, $\Box^{-1}\Gamma \vdash /[\Sigma]A$. By prop-consistencyfactsprop-consistencyfacts-b, $\Box^{-1}\Gamma \cup \{\neg A\}$ is Σ-consistent. By Lindenbaum's Lemma, there is a complete Σ-consistent set Δ such that $\Box^{-1}\Gamma \cup \{\neg A\} \subseteq \Delta$. By consistency, $A \notin \Delta$. $\Box$